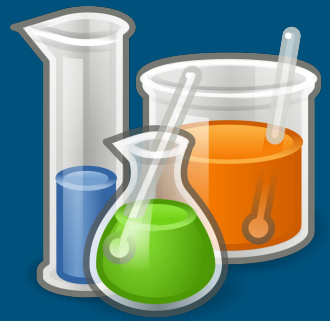
k-Nearest Neighbors (k-NN) Classification

Use on a selective subset of the Adult Income Dataset (1994)

k-NN Quick Summary

- Developed by Evelyn Fix and Joseph Hodges in 1951
- Used in supervised learning for classification and regression
- **In classification**: the output class is chosen from the most common class of the k nearest training data points around the input data point
- **In Regression**: the output value is the mean or center value of the k nearest training data points around the input data point
- This presentation is going to use k-NN for classification



Data Selection Goals

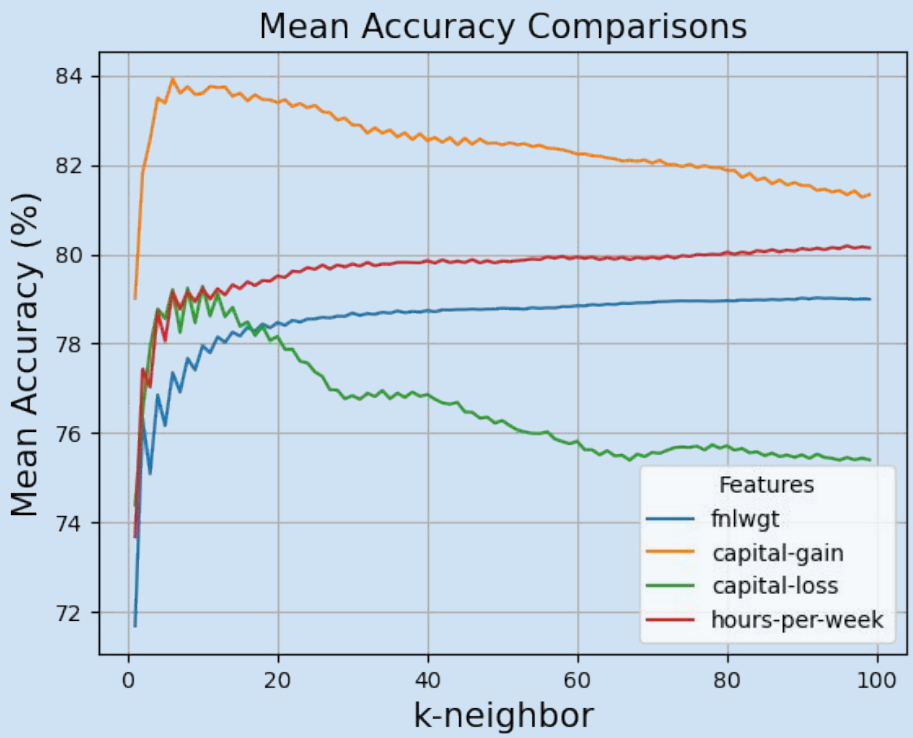
To achieve high accuracy of k-NN prediction, it's crucial to select our dataset from a number of data in the Adult Income dataset with the following goals in mind.

3rd Feature

- One of numerical features
  - fnlwgt
  - capital-gain
  - capital-loss
  - hours-per-week
- Highest accuracy of k-NN prediction on the overall data

Data

- No Duplicates
  - To reduce computation time and bias
- Same amount of data in both income categories (<=50K and >50K)
  - To reduce data to one income category would always make a 100% on prediction, but not useful and uninteresting.



k-NN prediction accuracy on the numerical features

- Input (X) features
  - age
  - education-num
  - (one of the selected features)
- Output (y) feature
  - Income
- k-neighbor ranges from 1 to 99
- Duplicates Removed
- Split: Train=80%, Test=20%
- Average prediction accuracy over 100 splits

“capital-gain” will be selected because it has the highest prediction accuracy overall.

Input Preprocessing

To improve training performance, some preprocessing need to be performed. In this case, StandardScaler algorithm will be chosen to remove the mean and scale to unit variance of the input. The standard score ***z*** of a sample ***x*** is calculated as:

$$z = (x - u) / s$$

where ***u*** is the mean of the training samples, and ***s*** is the standard deviation of the training samples.

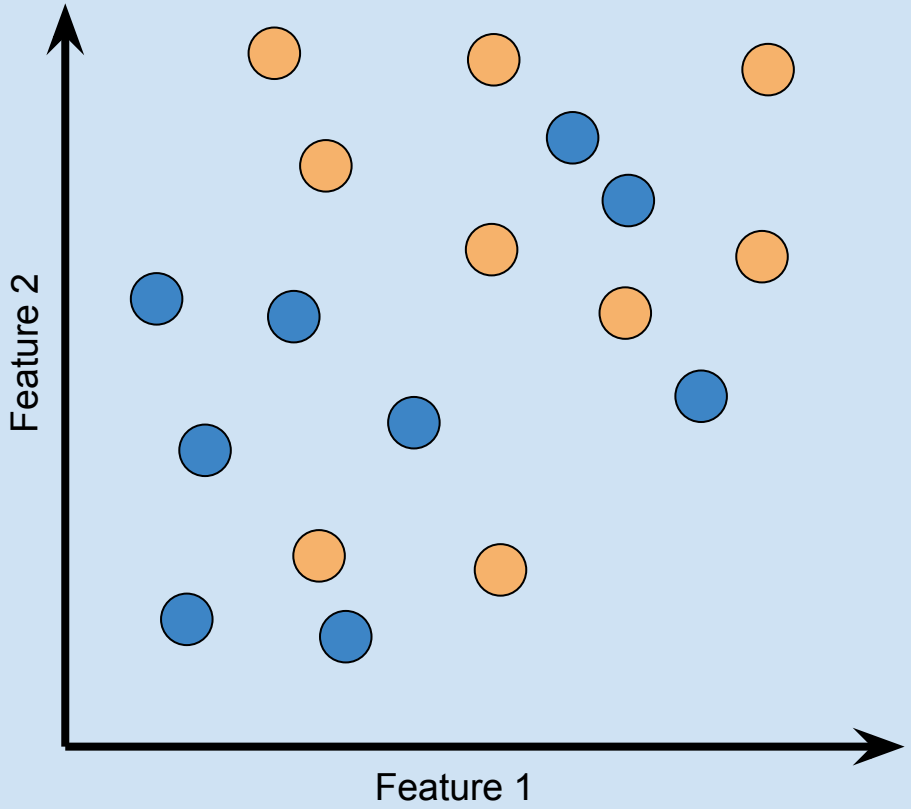
***Note that Output (y) will not be preprocessed because it does not have a numerical attribute.*

Data Point Removal Algorithm

Current prediction accuracy is about 83%. To improve, we need to systematically reduce the data points in the dataset using the following algorithm.

1. Find the N closest data points for every data point
2. Remove one data point from the dataset that is most closely surrounded by a different class
3. Compute the prediction accuracy of the rest of the data points
4. Repeat step 1 to 3 until the prediction accuracy reaches 100%

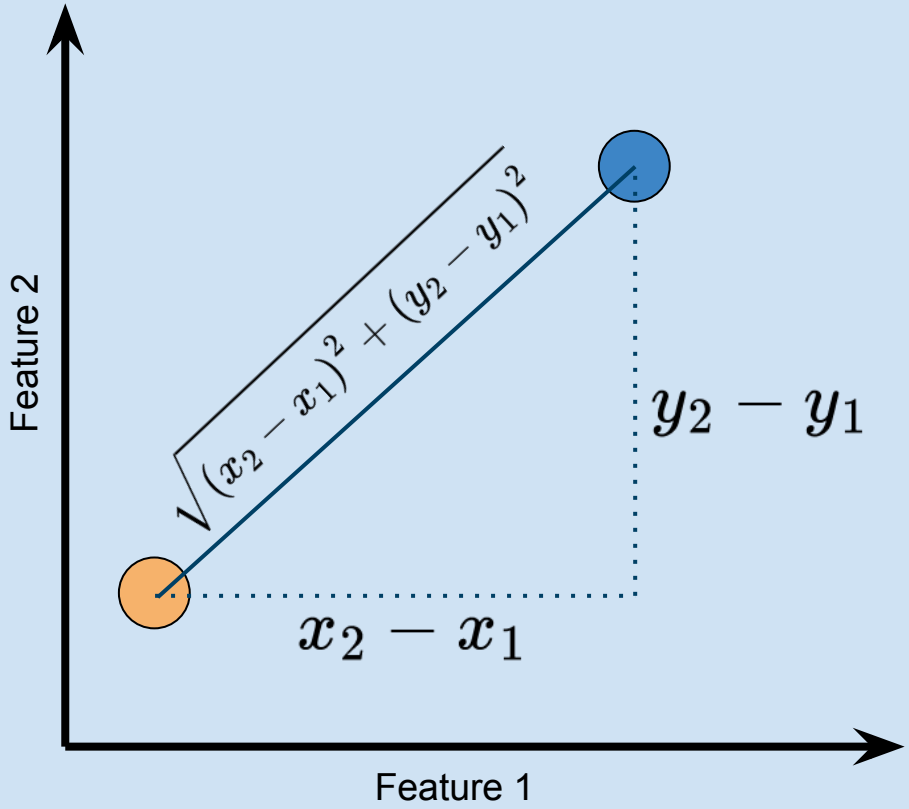
After the desired accuracy is reached, whatever is left will be our new dataset.



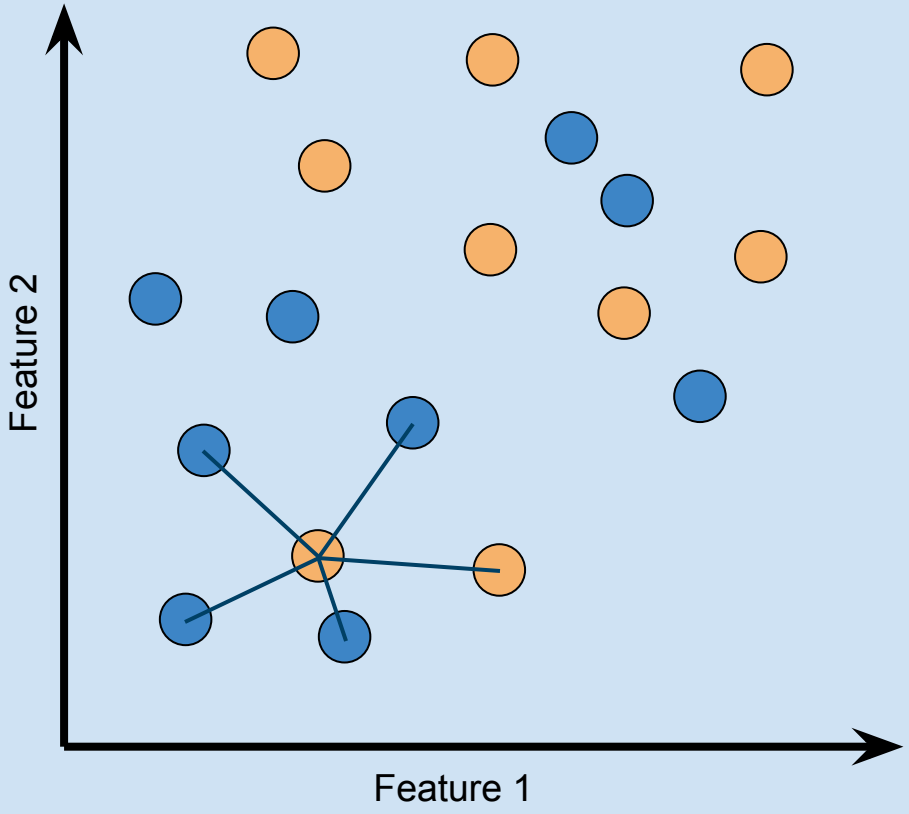
To demonstrate the algorithm, this example dataset has been created to have:

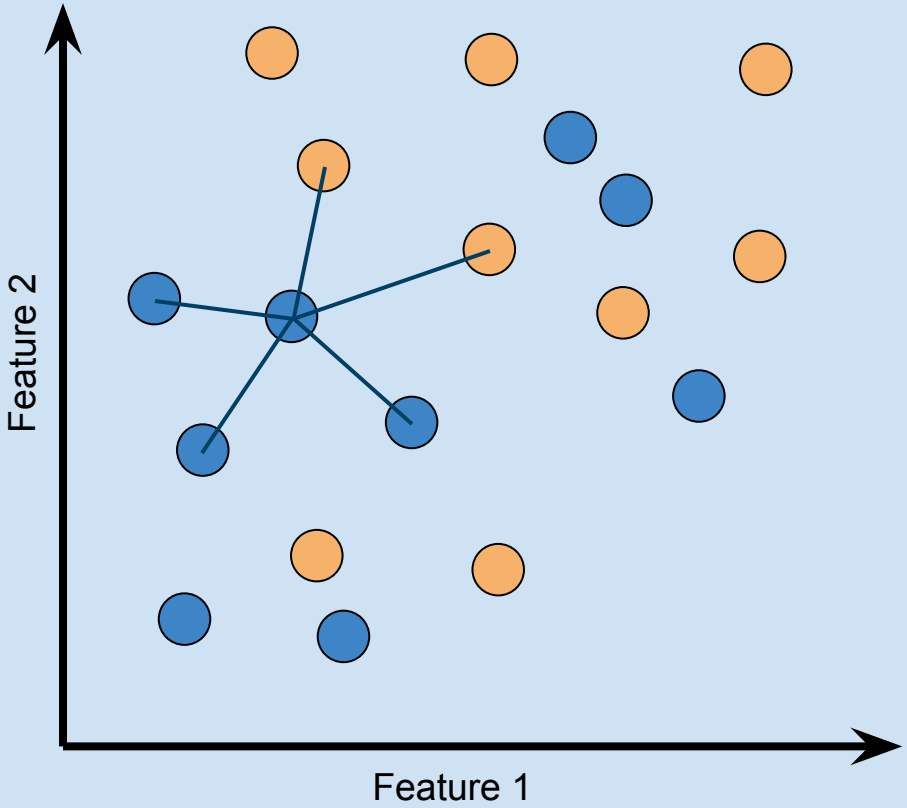
- 18 data points
- 2 numerical and 1 categorical features
- 2 classes for categorical feature
- 9 data points in each class

Even though it has 2 input features, this method can also be applied to dataset with more input features.

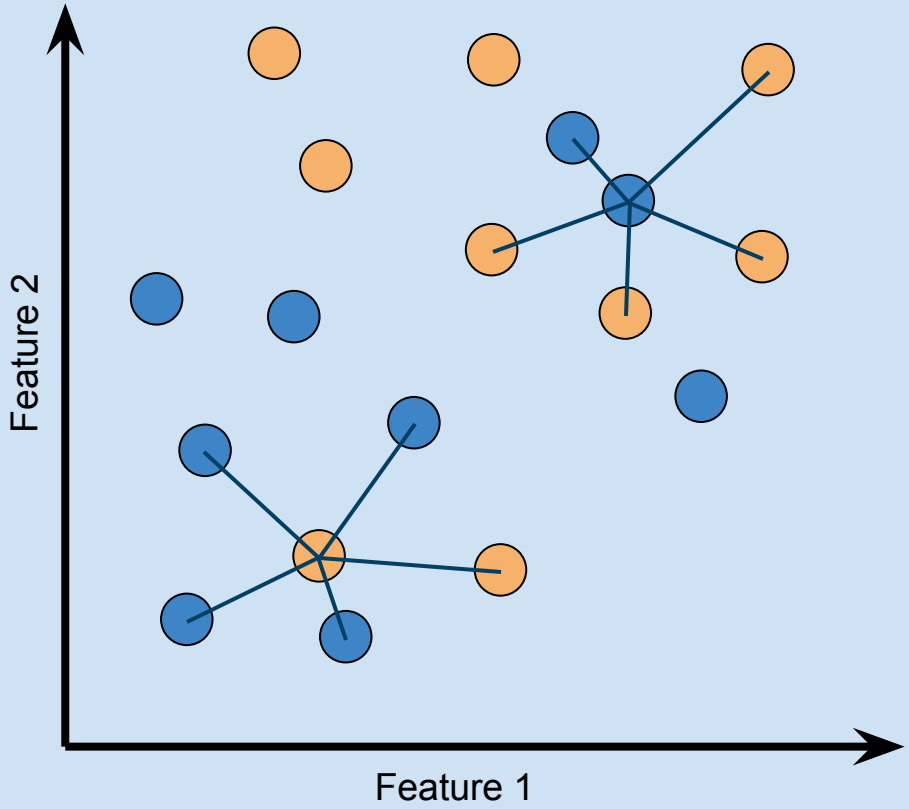


Find Euclidean distance of all data points by using Pythagorean theorem to calculate distance between a data point to other data points.





Find the N=5 closest data points for every data point
Remove one data point from the dataset that is most closely surrounded by a different class

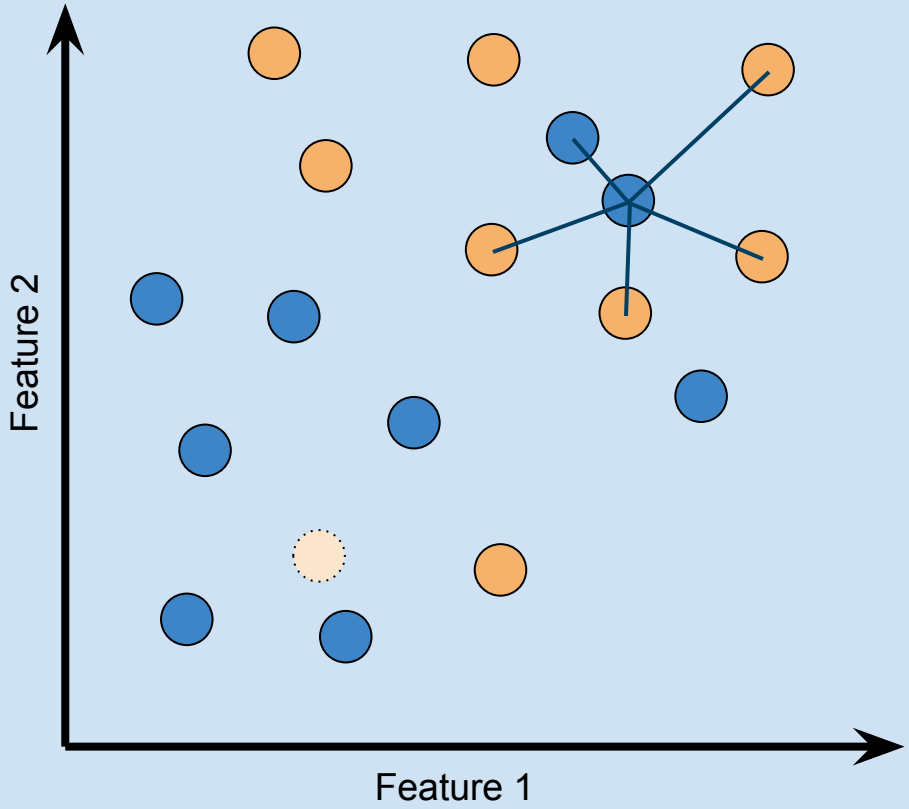


In case of two or more data points have the same amount of different class, It will be weighted by the order as described:

$$1 - \frac{i}{N}$$

Where:

- ***i*** is the order number in which the data point is closest to the target (start from 0, the closest)
- ***N*** is the number of closest data points picked for the calculation (in this example, N=5)



Conclusion

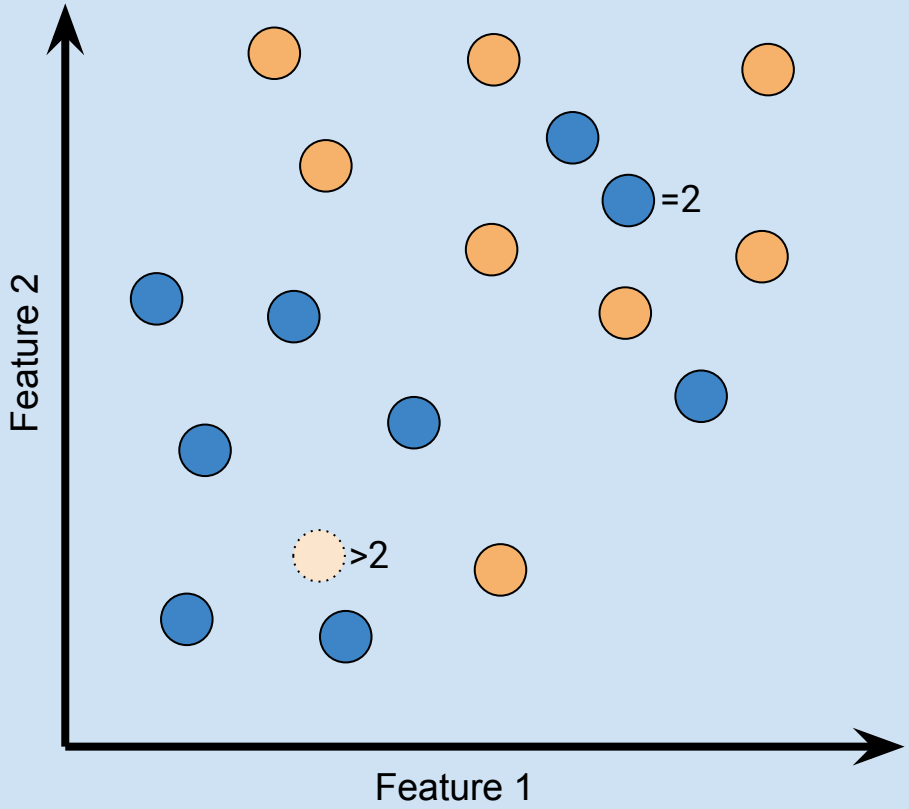
For each data point, find Euclidean distances between it and other data points then order them from least to most. After that, sum all values as described by:

$$\sum_{i=0}^{N-1}\left(1-\frac{i}{N}\right)C_i$$

Where:

- ***i*** and ***N*** are the same as previously described
- C_i is 0 if ***i***th closest data point has the same class as the target, and is 1, otherwise

After the calculation is performed on all data points, the one that has the most value will be removed.



If more than one data point have the same max value, pick one from the ordering of the dataset

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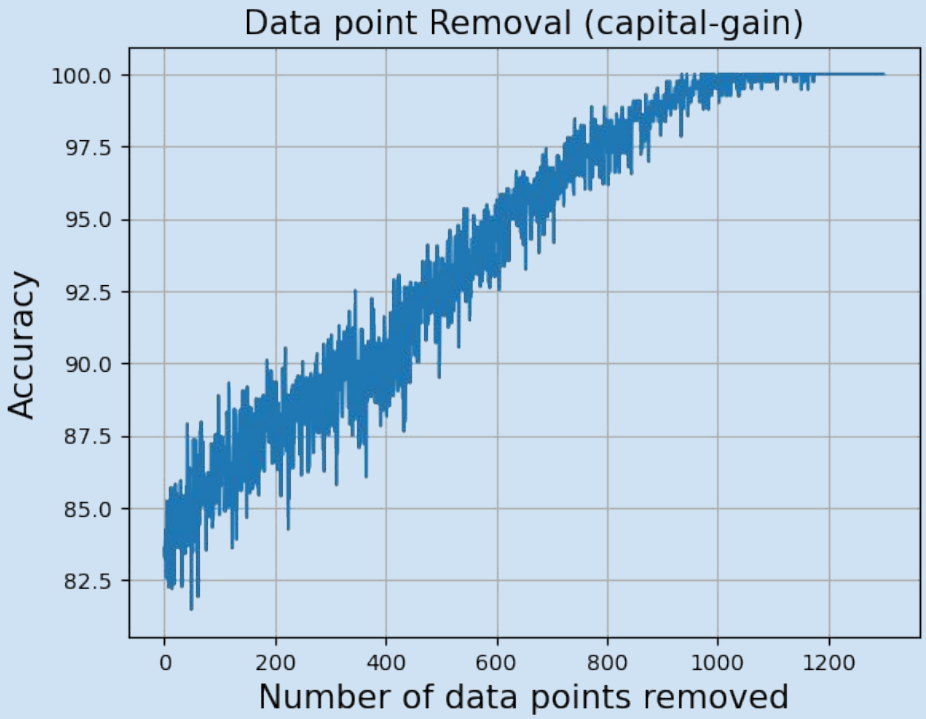
Notice that there are a lot of similarities between this and k-NN algorithm.

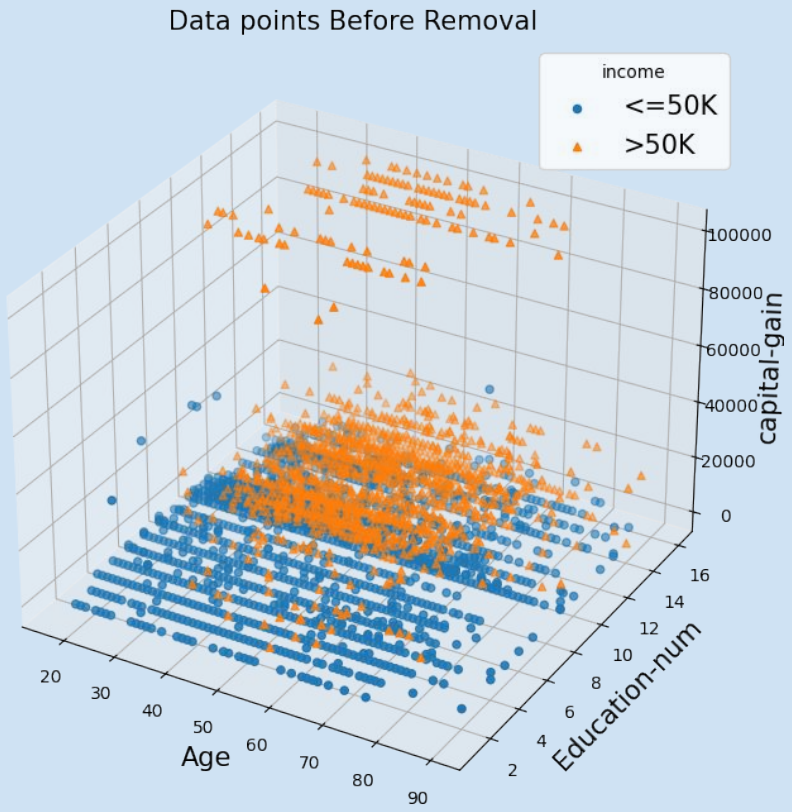


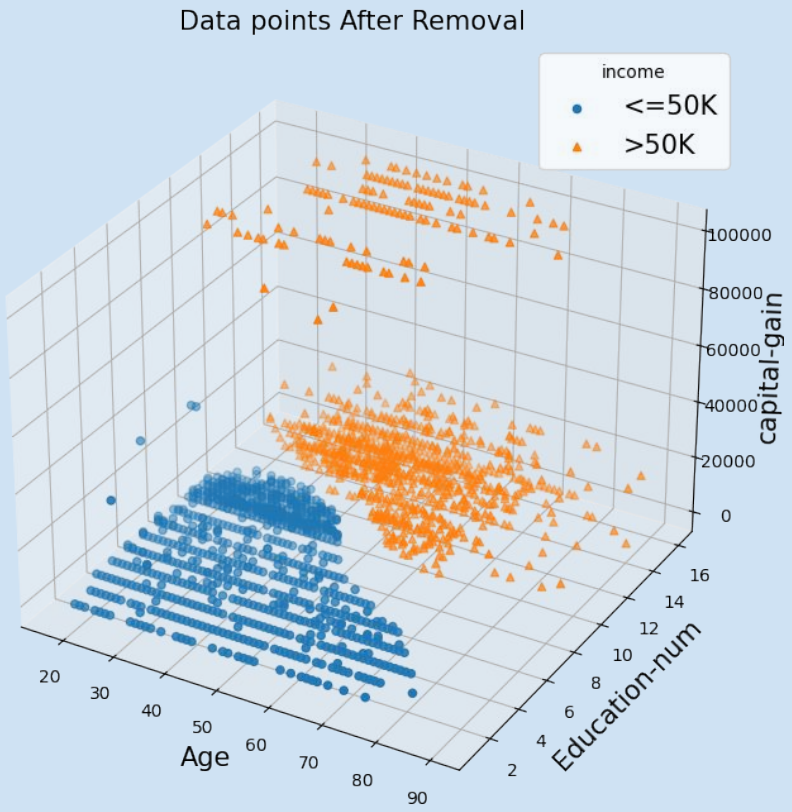
Data point removal of Adult dataset

After every single removal of a data point (N=500), calculate k-NN prediction accuracy on the rest of the data points using:

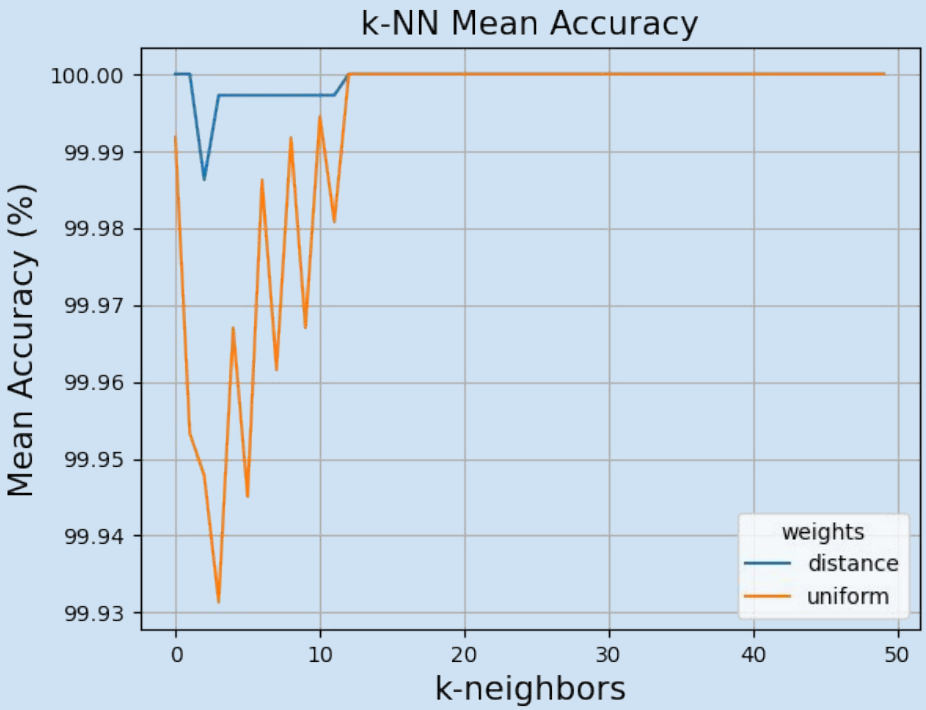
- Input (X) features
  - age
  - education-num
  - capital-gain
- Output (y) feature
  - Income
- k-neighbor = 15
- Split: Train=80%, Test=20%

Accuracy reaches 100% just before 1200 removals.





This shows a distinct separation between Before and After
Before: 1712+1307=3019, After: 910+910=1820



Final k-NN prediction accuracy on the new dataset

- Input (X) features
  - age
  - education-num
  - capital-gain
- Output (y) feature
  - Income
- k-neighbor ranges from 1 to 49
- Split: Train=80%, Test=20%
- Average prediction accuracy over 100 splits
- 2 weight functions:
  - Uniform: 1
  - Distance: 1/D (D = Euclidean distance)

This shows that the prediction accuracies are 100% for all k between 12 and 49

***This selective dataset has certainly caused the model to overfit*

The End

Thank you for your time